



# LABORATORY MANUAL

PROGRAMME: DIPLOMA

SEMESTER /YEAR : 3<sup>rd</sup> / 2<sup>nd</sup>

SUBJECT NAME: EDCI LAB

**JHARKHAND RAI UNIVERSITY KAMRE, RANCHI**  
**JHARKHAND**

| S.No | LIST OF EXPERIMENTS TO BE CONDUCTED                                 |
|------|---------------------------------------------------------------------|
| 1    | Forward and Reverse bias characteristics of PN junction diode       |
| 2    | To verify V-I characteristics of a Zener diode .                    |
| 3    | Set up the circuit and verify the waveforms of Half Wave Rectifier. |
| 4    | Set up the circuit and verify the waveforms of Full Wave Rectifier. |
| 5    | Input and Output characteristics of transistor in CE configuration  |
| 6    | Input and Output characteristics of transistor in CB configuration  |
| 7    | Input and Output characteristics of transistor in CC configuration  |
| 8    | FET characteristics                                                 |

**Aim:**

1. To plot the Volt – Ampere characteristics of given P – N junction Diode
2. To Find the Static, dynamic Forward and Reverse Resistance s of diode .

**Apparatus:**

1. Power supply
2. Diode
3. Resistance
4. Ammeter
5. Voltmeter
6. Bread board
7. Connecting Patch chords

**Procedure:****Forward Bias:**

- 1 . Connect the circuit as per the circuit diagram shown in

Fig (1) . For Both Silicon and Germanium Diodes .

- 2 . Vary the power supply in such a wa y that the readings are taken in steps of 0 . 1V, to the maximum reading of power supply of 30V .

- 3 . Note down the corresponding Ammeter and Voltmeter readings .

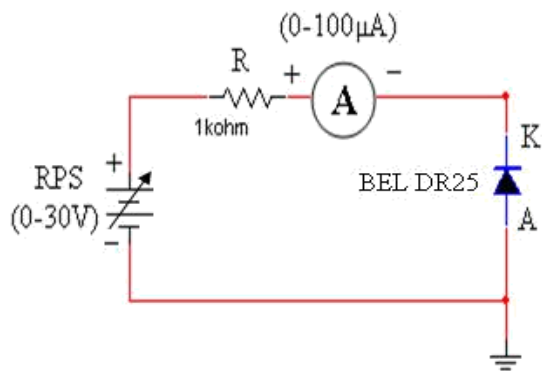
- 4 . Plot the graph between Forward voltage and current( $V_F$  and  $I_F$  . )

5 . Find the Static Forward Resistance  $R_F = V/I \Omega$  .

6 . Find the Dynamic Forward Resistance  $r_F = \frac{\Delta V}{\Delta I} \Omega$  .

### CIRCUIT DIAGRAM:

#### REVERSE BIAS:

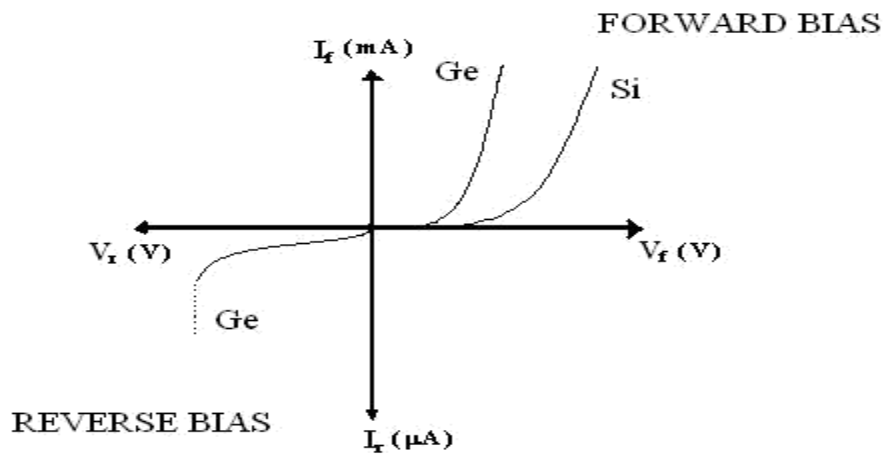


Fig(2)

### Tabular Column:

| S . No . | GERMANIUM          |                     |
|----------|--------------------|---------------------|
|          | V <sub>R</sub> (V) | I <sub>R</sub> (μA) |
|          |                    |                     |

## Model Graph:



## Reverse Bias:

1. Connect the circuit as per the circuit diagram shown in Fig(2).
2. Vary the power supply in such a way that the readings are taken in steps of 2V, to the maximum reading of power supply of 30V.
3. Note the corresponding Ammeter and Power Supply readings.
4. Plot the graph between  $V_R$  and  $I_R$ .
5. Find the Static Reverse Resistance  $R_r$
6. Find the Dynamic Reverse Resistance

## Precautions:

1. Don't give voltage to the circuit beyond prescribed range.
2. Don't short circuit the output terminal of power supply.
3. Carefully connect meter terminals (+ and -).

4 . Carefully connect p – n diode terminals (anode and cathode) .

### **Result:**

- 1 . Volt – Ampere Characteristics of P - N Diode are plotted .
- 2 . Static Forward Resistance  $R_F =$
- 3 . Static Reverse Resistance  $R_r =$
- 4 . Dynamic Forward Resistance =
- 5 . Dynamic Reverse Resistance =

### **CIRCUIT DIAGRAM:**

#### **FORWARD BIAS**

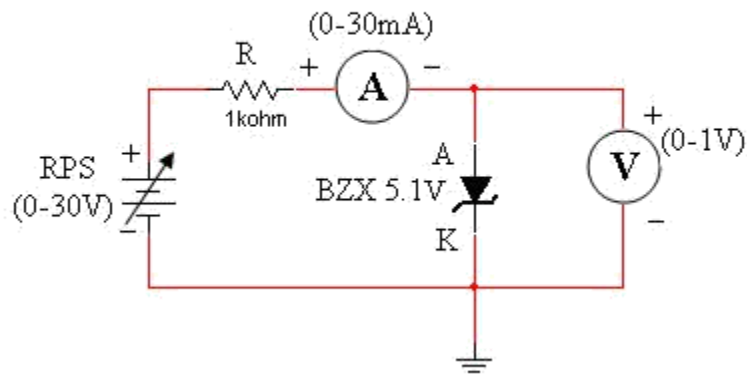


Fig (1)

**Tabular Column:**

**Forward Bias :**

| S . No . | $V_F$<br>(Volts) | $I_F$ (mA) |
|----------|------------------|------------|
|          |                  |            |

**Exp no :2**

## **ZENER DIODE CHARACTERISTICS**

### **Aim:**

- 1 . To Plot the Volt – Ampere characteristics of a Zener diode .
- 2 . To Find the Zener Break down Voltage, Dynamic Forward Resistance and Dynamic Reverse Resistance of Zener Diode .

### **Apparatus :**

- 1 . Power supply
- 2 . Zener Diode
- 3 . Ammeter
- 4 . Voltmeter
5. Resistor

### **Procedure:**

#### **Forward Bias:**

- 1 . Connect the circuit as per the circuit diagram shown in Fig(1) .
- 2 . Vary the power supply voltage in such a way that the readings are taken in steps of 0 . 1 V, to the maximum reading of power supply of 20 V .
- 3 . Note down the corresponding Ammeter and Voltmeter readings .
- 4 . Plot the graph between  $V_F$  and  $I_F$  .

5 . Find the dynamic forward resistance using

$$r_f = \frac{\Delta V}{\Delta I} \Omega$$

**CIRCUIT DIAGRAM:**

**FORWARD BIAS**

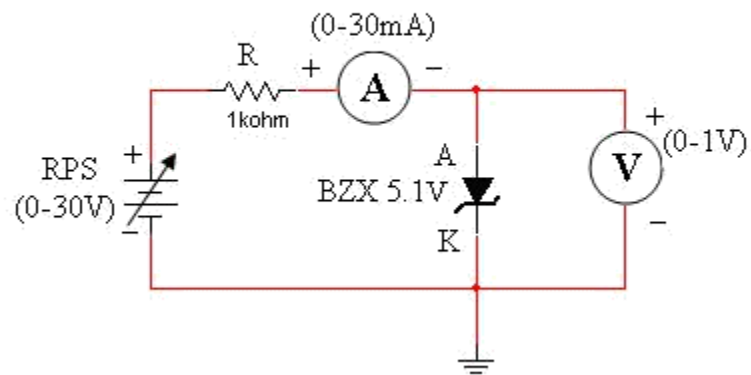


Fig (1)

### Tabular Column:

#### Forward Bias :

| S . No . | $V_F$<br>(Volts) | $I_F$ (mA) |
|----------|------------------|------------|
|          |                  |            |

### REVERSE BIAS

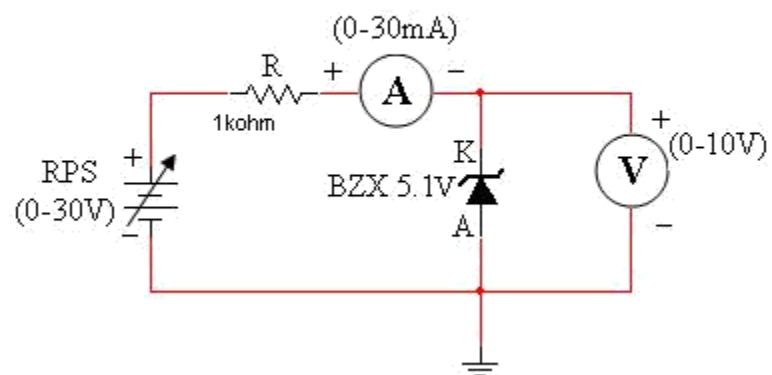


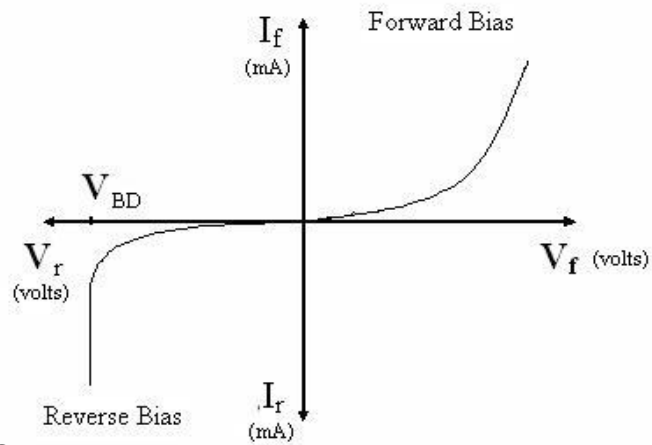
fig (2) .

**Tabular Column:**

**Reverse Bias :**

| S . No . | V <sub>R</sub> (Volts) | I <sub>R</sub> (mA) |
|----------|------------------------|---------------------|
|          |                        |                     |

**MODEL GRAPH:**



**Reverse Bias:**

- 1 . Connect the circuit as per the circuit diagram shown in

Fig (2) .

- 2 . Vary the power supply voltage in such a way that the readings are taken in steps of 2V, to the maximum reading of power supply of 20V
- 3 . Note down the corresponding Ammeter and Voltmeter Readings .
- 4 . Once break down occurs  $V_Z$  remains fairly constant even though  $I_Z$  increases .
- 5 . Plot the graph between  $V_R$  and  $I_R$  .
- 6 . Find the Dynamic Reverse Resistance  $r_R = \frac{\Delta V}{\Delta I} \Omega$

**Precautions:**

- 1 . Carefully connect the meter terminals (+ and – ) .
- 2 . Carefully connect the Zener diode terminals (Anode & Cathode) .

Exp no : 3

## HALF WAVE RECTIFIER

### Aim:

To Find the Ripple factor and Percentage of Regulation of a Half Wave Rectifier with out Filter.

### Apparatus:

- 1) Step down transformer
- 2) Diode
- 3) DRB (Decade Resistance Box)
- 4) Ammeter
- 5) Multimeter

### Procedure:

1. Connect the circuit as per the circuit diagram shown in Fig(1).
2. Note down the No Load DC Voltage  $V_{dc0}$  when  $I_{dc} = 0$
3. Vary the load resistance  $R_L$  (DRB) and note down  $I_{dc}$  and  $V_{dc}$ ,

$V_{ac}$

Using Multimeter

- a). Calculate the ripple factor ( r )

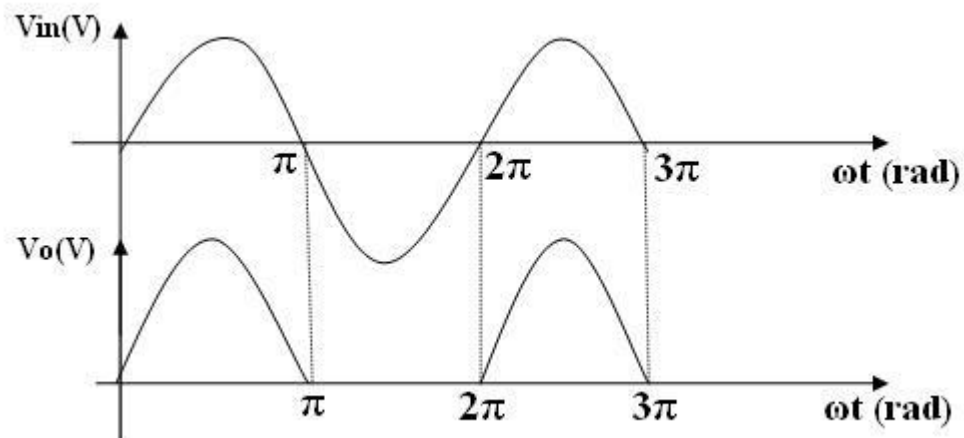
$$r = \frac{\text{RMS values of AC component}}{\text{Average value}} = \frac{V_{ac}}{V_{dc}}$$

- b). Calculate the Percentage of Regulation =  $\left[ \frac{(V_{dc0} - V_{dc})}{V_{dc}} \right] \times 100$  or  $\frac{(V_{NL} - V_{FL})}{V_{FL}} \times 100$

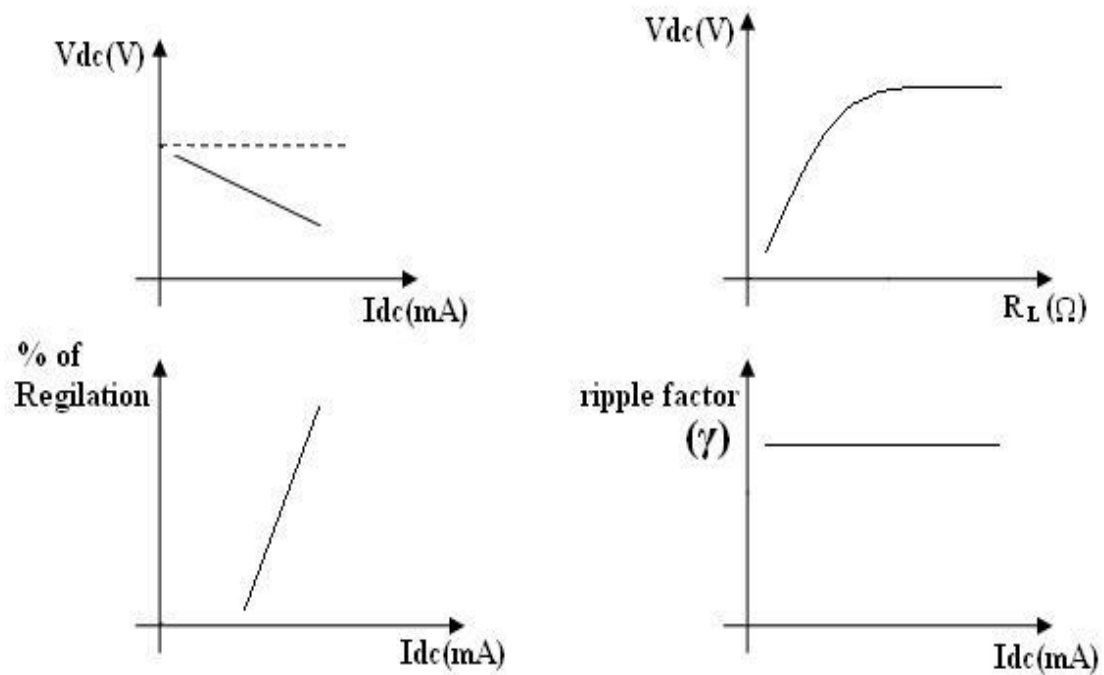
4. Draw the following graphs

- i).  $V_{dc}$  Vs  $R_L$
- ii)  $V_{dc}$  Vs  $I_{dc}$
- iii) Percentage of Regulation Vs  $I_{dc}$ . Taking  $V_{dc}$  on y – axis.

### MODEL WAVEFORMS:



### MODEL GRAPHS:



**Precautions:**

1. Don't give the voltage to the circuit beyond prescribed range.
2. Don't short circuit the output terminal.
3. Carefully connect meter & electrolytic capacitors terminals (+ and -)
4. Carefully connect diode terminals (anodes and cathodes).

**Result:**

Ripple factor and Percentage of Regulation of a Half Wave Rectifier with out Filter is found.  
Ripple Factor =  
Percentage of Regulation =

### CIRCUIT DIAGRAM

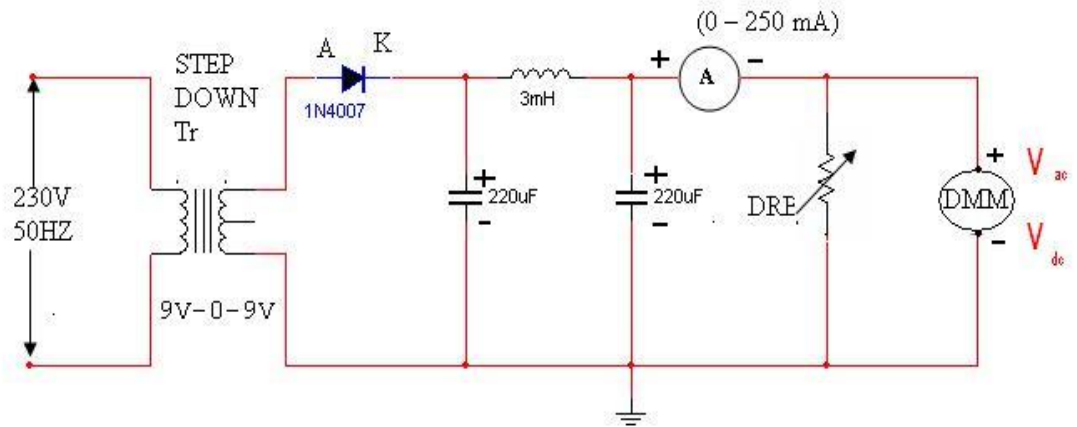


Fig (1)

### Tabular Column:

**No Load DC Voltage,  $V_{dc0}$  =**

| S.No. | RL<br>( $\Omega$ ) | I <sub>dc</sub><br>(mA) | V <sub>dc</sub><br>(V) | V <sub>ac</sub> (V) | r =<br>V <sub>ac</sub> /V <sub>dc</sub> | % of<br>Regulation =<br>[(V <sub>dco</sub> - V <sub>dc</sub> )/V <sub>dc</sub> ] X |
|-------|--------------------|-------------------------|------------------------|---------------------|-----------------------------------------|------------------------------------------------------------------------------------|
|       |                    |                         |                        |                     |                                         |                                                                                    |

# FULL WAVE RECTIFIER

## Aim:

To Find the Ripple factor and Percentage of Regulation of a Full Wave Rectifier without Filter.

## Apparatus:

- 1) Step down transformer
- 2) Diode
- 3) DRB (Decade Resistance Box)
- 4) Ammeter
- 5) Multimeter

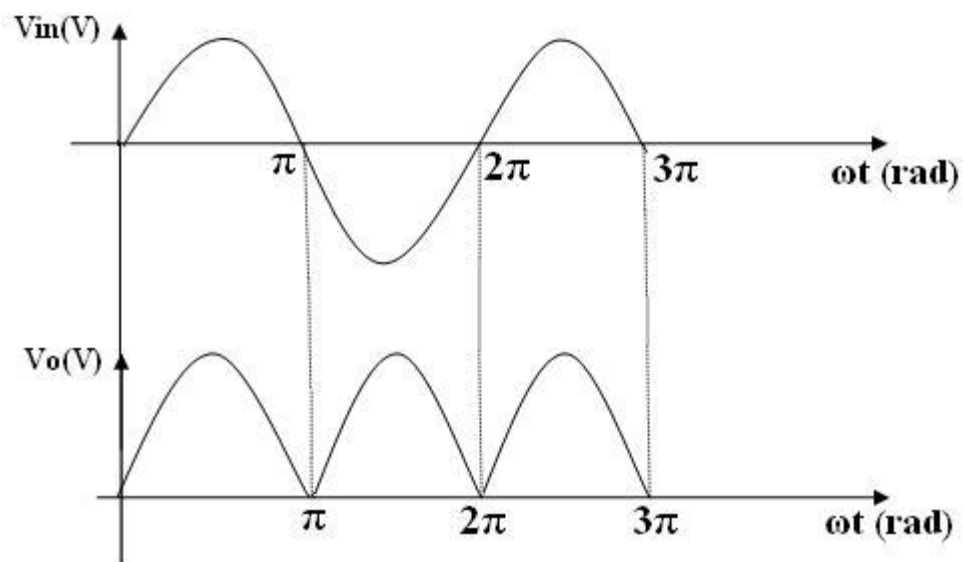
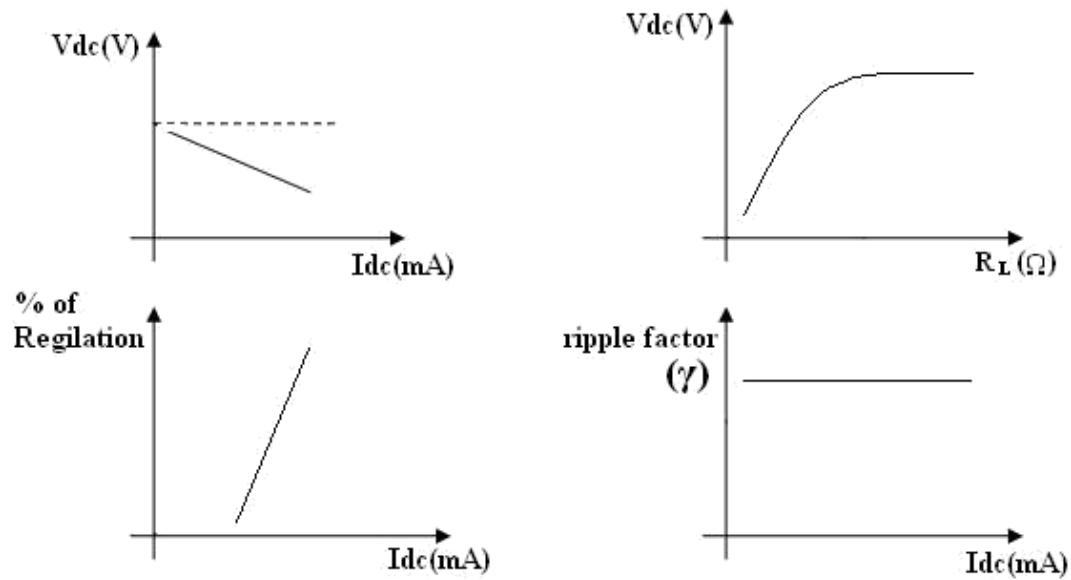
## Procedure:

1. Connect the circuit as per the circuit diagram shown in fig (1).
2. Note down the No Load DC Voltage  $V_{dc0}$  when  $I_{dc} = 0$
3. Vary the load resistance  $R_L$  (DRB) and note down  $I_{dc}$  and  $V_{dc}$ ,  $V_{ac}$  using Multimeter.
4. Calculate the ripple factor (  $r$  )  
$$r = \frac{\text{RMS values of AC component}}{\text{Average value}} = \frac{V_{ac}}{V_{dc}}$$
5. Calculate the Percentage of Regulation =  $[(V_{dc0} - V_{dc}) / V_{dc}] \times 100$   
( Or )  $(V_{NL} - V_{FL}) / V_{FL} \times 100$
6. Draw the following graphs
  - i).  $V_{dc}$  Vs  $R_L$

ii)  $V_{dc}$  Vs  $I_{dc}$

iii) Percentage of Regulation Vs  $I_{dc}$ . Taking  $V_{dc}$  on y – axis.

### MODEL GRAPH



**Precautions:**

1. Don't short circuit the output terminal.
2. Carefully connect meter & electrolytic capacitors terminals (+ and -)
3. Carefully connect diode terminals (anodes and cathodes).

**Result:**

Ripple factor and Percentage of Regulation of a Full Wave Rectifier with out Filter is found.  
Ripple Factor =  
Percentage of Regulation =

**CIRCUIT DIAGRAM:**

CIRCUIT DIAGRAM

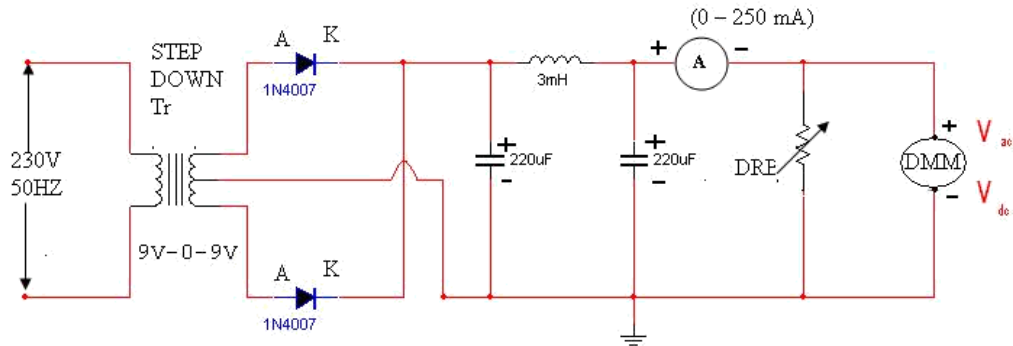


Fig (1)

**Tabular Column:**

No load D . C Voltage,  $V_{dc0} =$

| S.No. | $R_L$<br>( $\Omega$ ) | $I_{dc}$<br>(mA) | $V_{dc}$<br>(V) | $V_{ac}$ (V) | $r =$<br>$V_{ac}/V_{dc}$ | % of<br>Regulation =<br>$[(V_{dco} - V_{dc})/V_{dc}] \times$<br>100 |
|-------|-----------------------|------------------|-----------------|--------------|--------------------------|---------------------------------------------------------------------|
|       |                       |                  |                 |              |                          |                                                                     |

**Ex.No. 5**

### **CHARACTERISTICS OF BJT IN CE CONFIGURATION**

**AIM:** To study and plot the transistor characteristics in CE configuration.

#### **APPARATUS REQUIRED:**

Transistor Kit, Ammeter, Voltmeter,  
Resistor, Connecting Wires.

#### **THEORY:**

A BJT is a three terminal two – junction semiconductor device in which the conduction is due to both the charge carrier. Hence it is a bipolar device and it amplifies the sine waveform as they are transferred from input to output. BJT is classified into two types – NPN or PNP. A NPN transistor consists of two N types in between which a layer of P is sandwiched. The transistor consists of three terminal emitter, collector and base. The emitter layer is the source of the charge carriers and it is heavily doped with a moderate cross sectional area. The collector collects the charge carriers and hence moderate doping and large cross sectional area. The base region acts a path for the movement of the

charge carriers. In order to reduce the recombination of holes and electrons the base region is lightly doped and is of hollow cross sectional area. Normally the transistor operates with the EB junction forward biased.

In transistor, the current is same in both junctions, which indicates that there is a transfer of resistance between the two junctions. Hence known as transfer resistance of transistor.

### **PROCEDURE:**

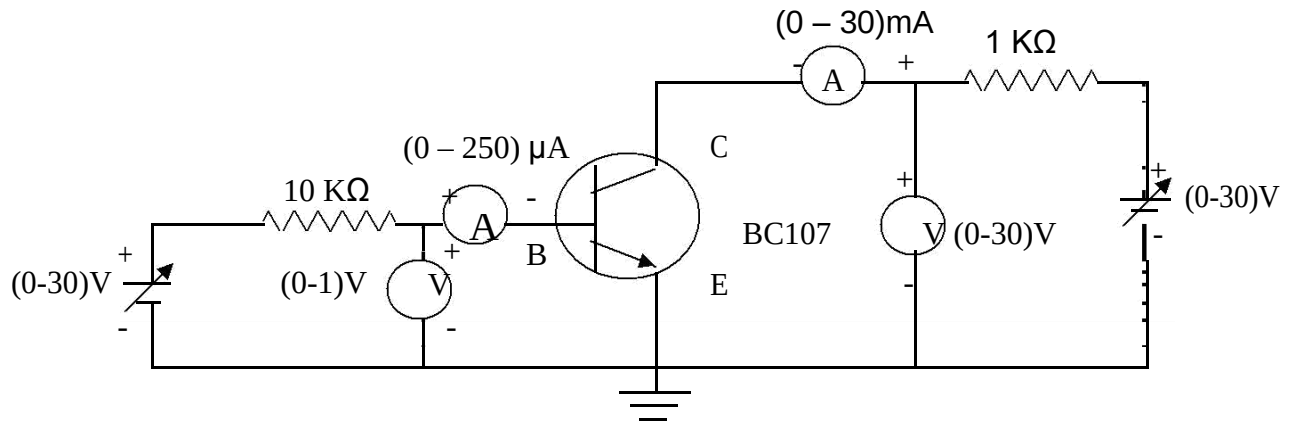
#### **INPUT CHARECTERISTICS:**

1. Connect the circuit as per the circuit diagram.
2. Set  $V_{CE}$  , vary  $V_{BE}$  in regular interval of steps and note down the Corresponding  $I_B$  reading. Repeat the above procedure for different values of  $V_{CE}$ .
3. Plot the graph:  $V_{BE}$  Vs  $I_B$  for a constant  $V_{CE}$ .

#### **OUTPUT CHARACTERISTICS:**

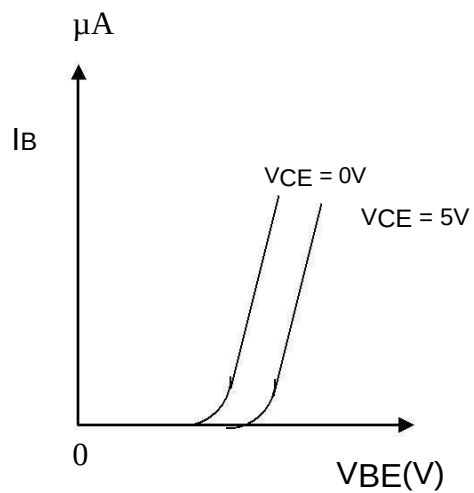
1. Connect the circuit as per the circuit diagram.
2. Set  $I_B$ , Vary  $V_{CE}$  in regular interval of steps and note down the corresponding  $I_C$  reading. Repeat the above procedure for different values of  $I_B$ .
3. Plot the graph:  $V_{CE}$  Vs  $I_C$  for a constant  $I_B$ .

### CIRCUIT DIAGRAM:

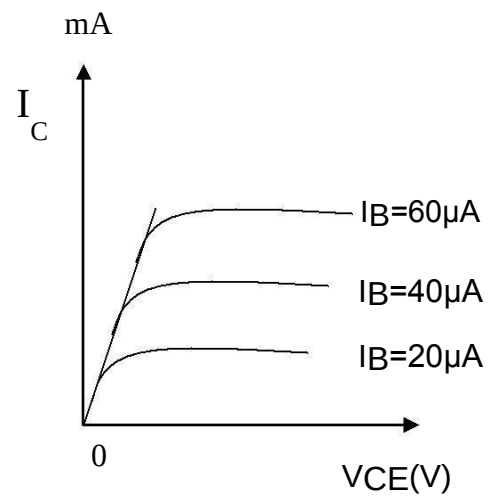


### MODEL GRAPH:

#### INPUT CHARACTERISTICS:



#### OUTPUT CHARACTERISTICS:



**TABULAR COLUMN:**

**INPUT CHARACTERISTICS:**

| VCE=1V |              | VCE=2V |              |
|--------|--------------|--------|--------------|
| VBE(V) | IB( $\mu$ A) | VBE(V) | IB( $\mu$ A) |
|        |              |        |              |

**OUTPUT CHARACTERISTICS:**

| IB=20 $\mu$ A |        | IB=40 $\mu$ A |        |
|---------------|--------|---------------|--------|
| VCE(V)        | IC(mA) | VCE(V)        | IC(mA) |
|               |        |               |        |

**RESULT:**

The transistor characteristics of a Common Emitter (CE) configuration were plotted

**Ex.No:6**

### **CHARACTERISTICS OF BJT IN CB CONFIGURATION**

**AIM:** To study and plot the transistor characteristics in CB configuration.

**APPARATUS REQUIRED:**

Transistor Kit, Ammeter, Voltmeter,  
Resistor, Connecting Wires.

**THEORY:**

In this configuration the base is made common to both the input and out. The emitter is given the input and the output is taken across the collector. The current gain of this configuration is less than unity. The voltage gain of CB configuration is high. Due to the high voltage gain, the power gain is also high. In CB configuration, Base is common to both input and output. In CB configuration the input characteristics relate  $I_E$  and  $V_{EB}$  for a constant  $V_{CB}$ . Initially let  $V_{CB} = 0$  then the input junction is equivalent to a forward biased diode and the characteristics resembles that of a diode. Where  $V_{CB} = +V_I$  (volts) due to early effect  $I_E$  increases and so the characteristics shifts to the left. The output characteristics relate  $I_C$  and  $V_{CB}$  for a constant  $I_E$ . Initially  $I_C$  increases and then it levels for a value  $I_C = \alpha I_E$ . When  $I_E$  is increased  $I_C$  also increases proportionality. Though increase in  $V_{CB}$  causes an increase in  $\alpha$ , since  $\alpha$  is a fraction, it is negligible and so  $I_C$  remains a constant for all values of  $V_{CB}$  once it levels off.

**PROCEDURE:****INPUT CHARACTERISTICS:**

It is the curve between emitter current  $I_E$  and emitter-base voltage  $V_{BE}$  at constant collector-base voltage  $V_{CB}$ .

1. Connect the circuit as per the circuit diagram.
2. Set  $V_{CE}=5V$ , vary  $V_{BE}$  in steps of  $0.1V$  and note down the corresponding  $I_B$ . Repeat the above procedure for  $10V$ ,  $15V$ .
3. Plot the graph  $V_{BE}$  Vs  $I_B$  for a constant  $V_{CE}$ .
4. Find the  $h$  parameters.

**OUTPUT CHARACTERISTICS:**

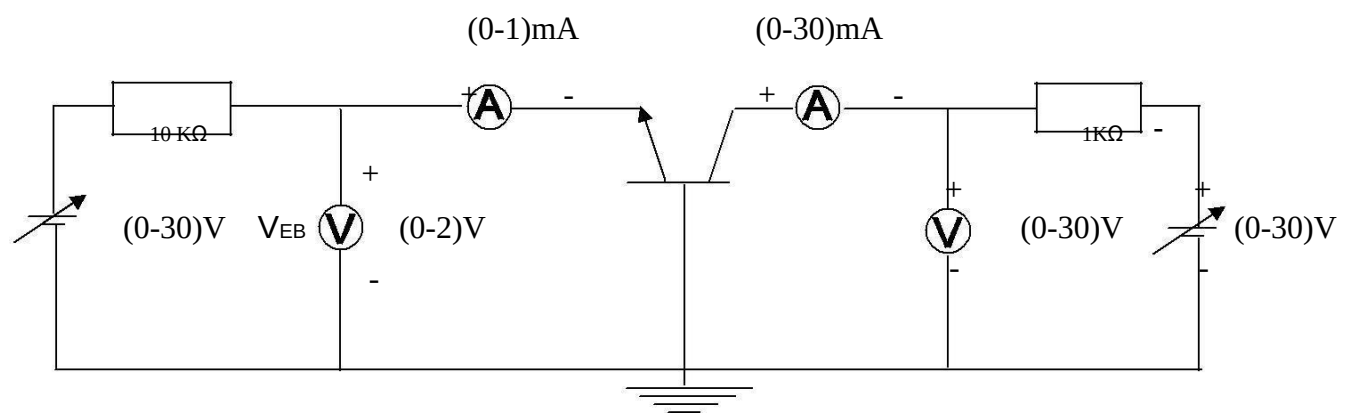
It is the curve between collector current  $I_C$  and collector-base voltage  $V_{CB}$  at constant emitter current  $I_E$ .

1. Connect the circuit as per the circuit diagram.
2. Set  $I_B=20\mu A$ , vary  $V_{CE}$  in steps of  $1V$  and note down the corresponding  $I_C$ .  
Repeat the above procedure for  $40\mu A$ ,  $80\mu A$ , etc.
3. Plot the graph  $V_{CE}$  Vs  $I_C$  for a constant  $I_B$ .
4. Find the  $h$  parameters

**RESULT:**

The transistor characteristics of a Common Base (CB) configuration were plotted and studied.

**CIRCUIT DIAGRAM:**



**TABULAR COLUMN:**

**INPUT CHARACTERISTICS:**

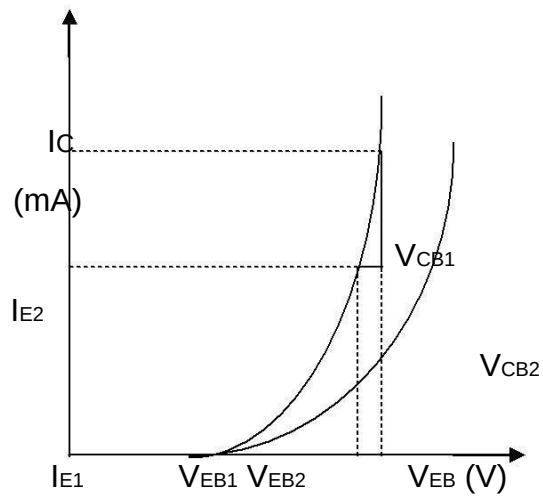
| S.No. | V <sub>CB</sub> =        V |                              | V <sub>CB</sub> =        V |                              | V <sub>CB</sub> =        V |                              |
|-------|----------------------------|------------------------------|----------------------------|------------------------------|----------------------------|------------------------------|
|       | V <sub>EB</sub><br>(V)     | I <sub>E</sub><br>( $\mu$ A) | V <sub>EB</sub><br>(V)     | I <sub>E</sub><br>( $\mu$ A) | V <sub>EB</sub><br>(V)     | I <sub>E</sub><br>( $\mu$ A) |
|       |                            |                              |                            |                              |                            |                              |

**OUTPUT CHARACTERISTICS:**

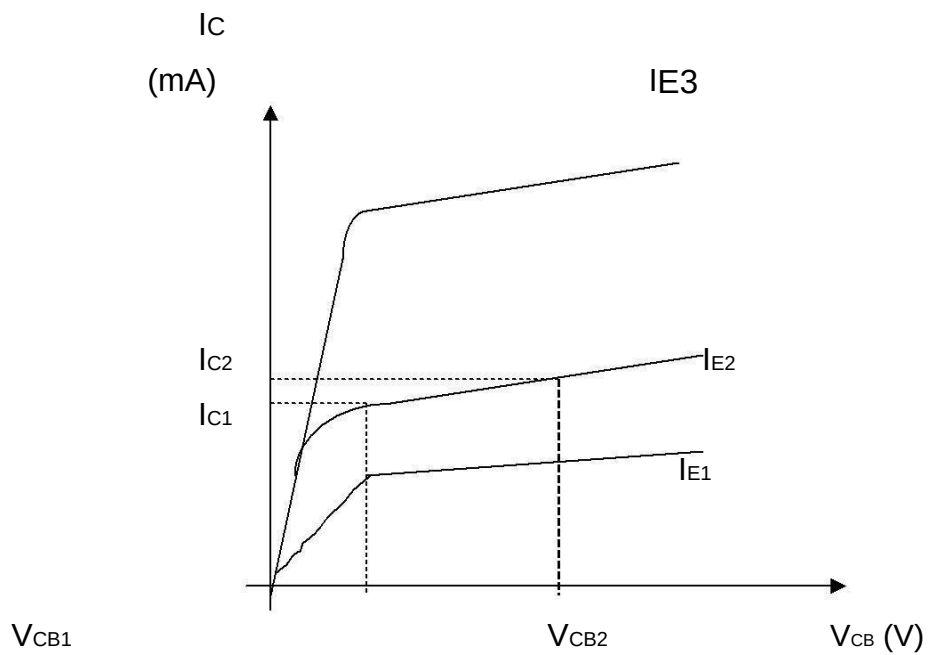
| S.No. | I <sub>E</sub> =        mA |                        | I <sub>E</sub> =        mA |                        | I <sub>E</sub> =        mA |                        |
|-------|----------------------------|------------------------|----------------------------|------------------------|----------------------------|------------------------|
|       | V <sub>CB</sub><br>(V)     | I <sub>C</sub><br>(mA) | V <sub>CB</sub><br>(V)     | I <sub>C</sub><br>(mA) | V <sub>CB</sub><br>(V)     | I <sub>C</sub><br>(mA) |
|       |                            |                        |                            |                        |                            |                        |

### MODEL GRAPH:

#### INPUT CHARACTERISTICS:



#### OUTPUT CHARACTERISTICS:



**Ex.No:7**

## **CHARACTERISTICS OF BJT IN CC CONFIGURATION**

**AIM:** To plot and study the transistor characteristics in CC configuration.

**APPARATUS REQUIRED:** Transistor Kit, Ammeter, Voltmeter, Resistor, Connecting Wires.

### **THEORY**

A BJT is a three terminal two – junction semiconductor device in which the conduction is due to both the charge carrier. Hence it is a bipolar device and it amplifies the sine waveform as they are transferred from input to output. BJT is classified into two types – NPN or PNP. A NPN transistor consists of two N types in between which a layer of P is sandwiched. The transistor consists of three terminal emitter, collector and base. The emitter layer is the source of the charge carriers and it is heavily doped with a moderate cross sectional area.

The collector collects the charge carriers and hence moderate doping and large cross sectional area. The base region acts a path for the movement of the charge carriers. In order to reduce the recombination of holes and electrons the base region is lightly doped and is of small cross sectional area. Normally the transistor operates with the EB junction forward biased.

In transistor, the current is same in both junctions, which indicates that there is a transfer of resistance between the two junctions. One to this fact the transistor is known as transfer resistance of transistor.

**PROCEDURE:**

**INPUT CHARECTERISTICS:**

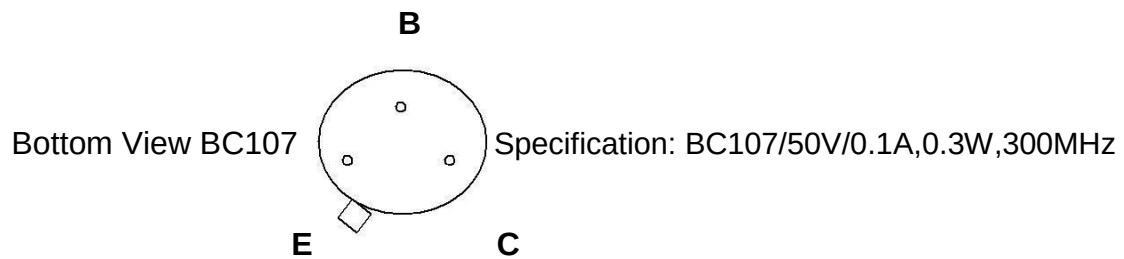
1. Connect the circuit as per the circuit diagram.
2. Set  $V_{CE}$ , vary  $V_{BE}$  in regular interval of steps and note down the corresponding  $I_B$  reading. Repeat the above procedure for different values of  $V_{CE}$ .
3. Plot the graph:  $V_{BE}$  Vs  $I_B$  for a constant  $V_{CE}$ .

**OUTPUT CHARECTERISTICS:**

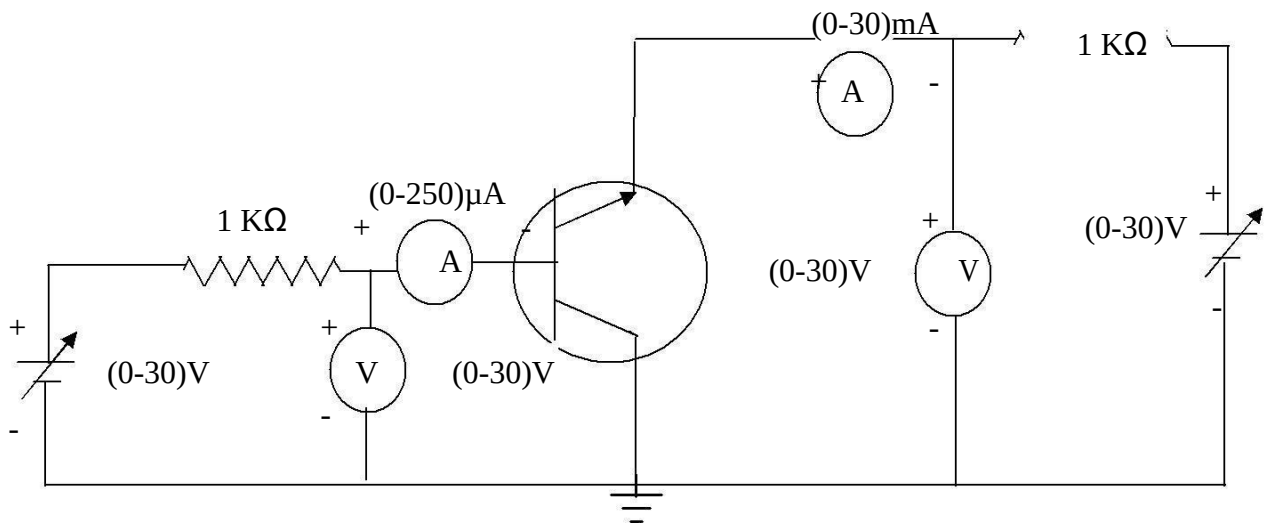
1. Connect the circuit as per the circuit diagram.
2. Set  $I_B$ , Vary  $V_{CE}$  in regular interval of steps and note down the corresponding  $I_C$  reading. Repeat the above procedure for different values of  $I_B$ .
3. Plot the graph:  $V_{CE}$  Vs  $I_C$  for a constant  $I_B$ .

**RESULT:**

The transistor characteristics of a Common Emitter (CC) configuration were plotted.

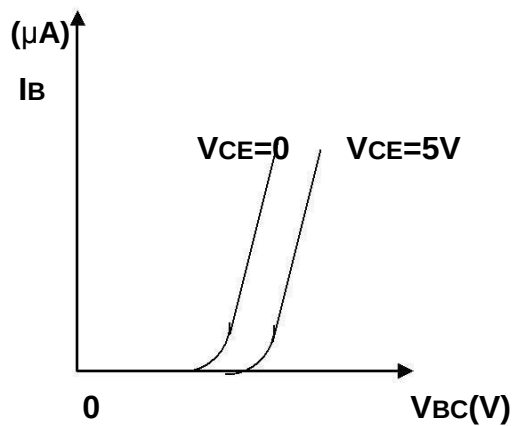


### CIRCUIT DIAGRAM:

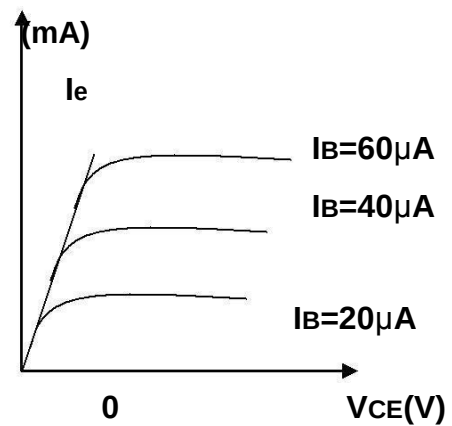


### MODEL GRAPH:

#### INPUT CHARACTERISTICS:



#### OUTPUT CHARACTERISTICS:



**TABULAR COLUMN:**

**INPUT CHARACTERISTICS:**

| <b>VCE=1V</b> |                              | <b>VCE=2V</b> |                              |
|---------------|------------------------------|---------------|------------------------------|
| <b>VBC(V)</b> | <b>IB(<math>\mu</math>A)</b> | <b>VBC(V)</b> | <b>IB(<math>\mu</math>A)</b> |
|               |                              |               |                              |

**OUTPUT CHARACTERISTICS:**

| <b>IB=20<math>\mu</math>A</b> |               | <b>IB=40<math>\mu</math>A</b> |               |
|-------------------------------|---------------|-------------------------------|---------------|
| <b>VCE(V)</b>                 | <b>IE(mA)</b> | <b>VCE(V)</b>                 | <b>IE(mA)</b> |
|                               |               |                               |               |

**Exp no : 8**

## **FET CHARACTERISTICS**

### **Aim:**

- 1 . To Obtain Drain and Transfer characteristics of FET connected Common Source configuration .
- 2 . To Obtain  $r_d$  ,  $g_m$  and  $\mu$  of FET .

### **Apparatus:**

|                    |            |         |
|--------------------|------------|---------|
| 1 . Power supplies | 0 – 30 V   | – 1 No  |
| 2 . FET            | BFW 10     | – 1 No  |
| 3 . Voltmeter      | (0 – 10 V) | – 2 Nos |
| 4 . Ammeter        | 0 – 30 mA  | – 1 No  |

### **Procedure:**

#### **Transfer characteristics:**

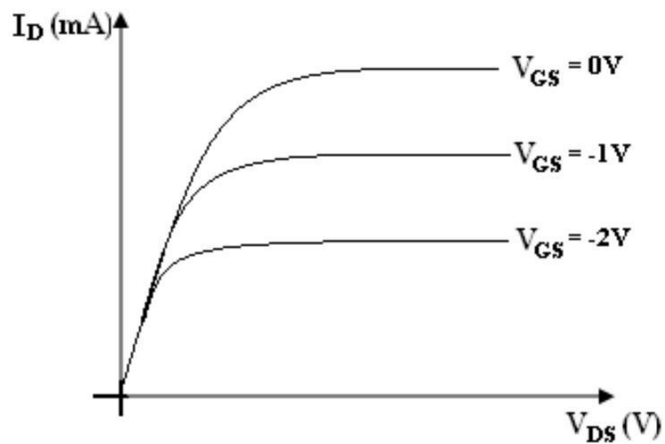
- 1 . Connect the circuit as per the circuit diagram shown in Fig (1) .
- 2 . Set  $V_{DS} = 2V$  by adjusting  $V_{DD}$  .
- 3 . Vary the input voltage  $V_{GS}$  and note the readings of  $I_D$  and  $V_{GS}$  .
- 4 . Repeat the above procedure for  $V_{DS} = 4V$  .
- 5 . Plot the transfer characteristics  $V_{GS}$  Vs  $I_D$  for constant values of  $V_{DS} = 2V$  and  $4V$  .
- 6 . Find Trans Conductance  $g_m = \frac{\Delta I_D}{\Delta V_{GS}} \bigg/ V_{DS} = \text{Constant}$

**Output (or) drain characteristics:**

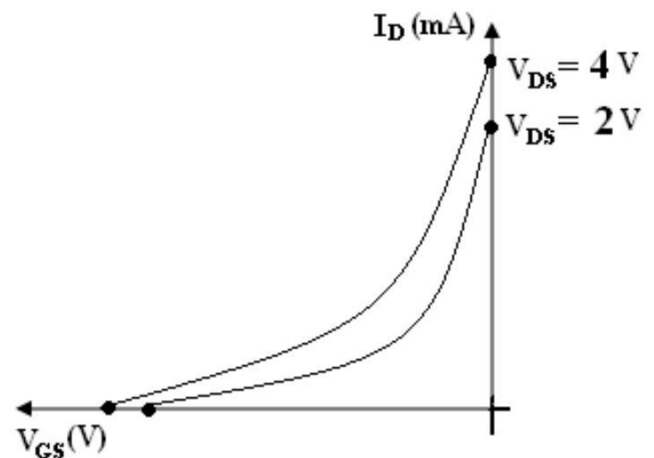
| S. No | $V_{GS} = 0V$ |            | $V_{GS} = -1V$ |            | $V_{GS} = -2V$ |            |
|-------|---------------|------------|----------------|------------|----------------|------------|
|       | $V_{DS} (V)$  | $I_D (mA)$ | $V_{DS} (V)$   | $I_D (mA)$ | $V_{DS} (V)$   | $I_D (mA)$ |
|       |               |            |                |            |                |            |

### Model graphs:

### Output (or) drain characteristics



**Transfer characteristics:**



### **Output (or) drain characteristics:**

- 1 . Connect the circuit as per the circuit diagram shown in Fig (1) .
- 2 . Set  $V_{GS} = 0V$  by adjusting  $V_{GG}$  .
- 3 . Vary the supply voltage  $V_{DD}$  and note the readings of  $I_D$  And  $V_{DS}$  . Take  $V_{DS} = V_{DD}$  .
- 4 . Repeat the above procedure for  $V_{GS} = -1V$  and  $-2V$  .
- 5 . Plot the output characteristics  $V_{DS}$  Vs  $I_D$  for constant values of  $V_{GS} = 0V, -1V$  and  $-2V$  .
- 6 . Find Drain Resistance  $r_d = \frac{\Delta V_{DS}}{\Delta I_D}$   $V_{GS} = \text{Constant and}$   
Amplification Factor,  $\mu = g_m \times r_d$

### **Precautions:**

1. Don't short circuit the output terminal
2. Don't give the voltage to the circuit beyond the prescribed range
3. Carefully vary the power supply
4. Carefully connect the FET terminals

### **Result:**

Drain and Transfer characteristics of FET Connected in Common Source configuration are obtained .

Trans Conductance,  $g_m =$

Drain Resistance,  $r_d =$

Amplification Factor,  $\mu = r_d * g_m$